

Smart Animal Feed Production Machine Using Arduino

Project Report

Abstract

The Smart Animal Feed Production Machine Using Arduino is an embedded systems prototype designed to convert organic kitchen and vegetable waste into animal feed through automated shredding, additive dispensing, and mixing. It integrates an Arduino UNO, L298N motor driver, servo motor, DC geared motors, LED, buzzer, and push-button control.

Introduction

Organic waste from households and vegetable markets is often discarded, causing pollution. This project demonstrates a low-cost automated solution that converts suitable organic waste into animal feed.

Objectives

Reduce organic waste, automate feed preparation, minimize manual effort, and demonstrate embedded system automation.

Hardware Components

Arduino UNO, L298N Motor Driver, SG90 Servo Motor, 160 RPM DC Geared Motor, TT Motor, Push Button, LED, Buzzer, Breadboard, Jumper Wires, 9V Battery.

Working Principle

Pressing the push button starts the sequence. The LED and buzzer indicate startup. The servo dispenses additives, the shredding motor processes waste, the mixing motor blends the material, and the system signals completion before returning to standby.

Applications

Small farms, poultry farms, waste recycling, smart agriculture, and educational embedded systems projects.

Advantages

Low-cost, modular, automated, environmentally friendly, and suitable for prototype-based learning.

Future Enhancements

Load cell and HX711 integration, LCD display, IoT connectivity, moisture sensing, mobile application, and solar-powered operation.

Conclusion

The prototype demonstrates the feasibility of automating organic waste processing into animal feed using affordable embedded hardware.